

ENGINEERING RECORD

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# Software Architecture Record

The system as verified against `main` — every claim about the present checked, every claim about the future named with its issue.

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keel · CodeGateSoftware/keel · verified 2026-08-31 · companions: docs/evolution-plan.md, docs/decisions/

**Status:** living document. Facts below verified against `main` on 2026-08-31 (through the #649 merge). Where a claim is about the future, it names the issue that carries it. **Companions:** `docs/evolution-plan.md` (where this architecture is going), `docs/decisions/` (the ADRs — individual load-bearing choices), the runbooks (`docs/operator-runbook.md`, `docs/go-live-runbook.md`).

## 1. What keel is

A self-hosted trading engine whose defining property is that its evidence discipline is part of the architecture, not a policy layered on top: a significance tool that cannot say no is refused at review; scores report and gate but never rank; negative results are published. Every architectural decision below is tested against that constitution.

It ships today in three shapes, all from one artifact set:

- **CLI** — `keel` commands over a local database (the operator's terminal).
- **Local web console** — `keel serve`, loopback-only, zero-dependency static UI.
- **Desktop distribution** — unsigned installers plus the `keel update` self-updater.

## 2. Package and module architecture

Monorepo, one distribution per concern, Python floor 3.14 (`requires-python = ">=3.14"`):

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| keel-trader (root)             | the application package                                      |
| └─ keel/agent.py               | the cadence: fires hourly, trades once per UTC day           |
| └─ keel/cli.py                 | command surface                                              |
| └─ keel/commands/              | console modules – one concern per file (research, confirm,   |
|                                | update, serve, brokers, ...); thinness is pinned by test     |
| └─ keel/strategy/              | rules, backtest engine, exit policy, CTS indicators,         |
|                                | paper trading, the promotion ladder                          |
| └─ keel/research/              | thirteen evidence modules (significance, montecarlo, cscv,   |
|                                | deflate, walkforward, independence, throughput, cts_factors, |
|                                | ledger, pooled_review, bias, matrix, tuning)                 |
| └─ keel/analysis/              | candles, indicators, levels, P&L, regime                     |
| └─ keel/data/                  | db, history, market_feed, gaps, freshness, cb_client         |
| └─ keel/execution/             | order placement and the fill path                            |
| └─ keel/compliance/            | screen.py (the attestation ceremony), purification.py        |
| └─ keel/sim/                   | simulation                                                   |
| └─ keel/web/                   | server, api, payload, events (SSE), staticfiles, security    |
| └─ keel/mcp/                   | the read-only MCP server (query_only enforced)               |
| └─ keel/templates/             | config templates                                             |
|                                |                                                              |
| packages/keel-core             | shared kernel                                                |
| packages/keel-broker-api       | the Broker port + capabilities contract                      |
| packages/keel-broker-coinbase  | live venue                                                   |
| packages/keel-broker-alpaca    | equities venue (cash-only rail enforced)                     |
| packages/keel-broker-kraken    | port-complete stub, honest about implementing nothing (#313) |
| packages/keel-broker-robinhood | dev-only; refuses to fake missing wiring                     |
| packages/keel-broker-fake      | the test venue                                               |

The stub pattern above is architecture, not placeholder: an adapter that returned empty lists would *look* working while answering every question with nothing. Refusal to fake is a load-bearing property across the codebase.

### 3. The load-bearing decisions, and what each carries

| Decision (in the code today)                                                           | Carries                                                                                                         |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Wheel-per-concern; same artifacts for CLI, desktop, hosted                             | Free and paid tiers are literally the same bytes; the evolution's hosted tier needs no engine fork              |
| Single-user engine + SQLite WAL per profile (ADR 0002)                                 | Container-per-tenant hosting with zero engine rewrite; no phase requires in-process multi-tenancy               |
| Broker port + capabilities (incl. the <code>multiplier == 1</code> cash rail, #372)    | Venue breadth as extension, not surgery. The rail that blocks options (#637) is the product working as designed |
| Compliance as ceremony + data ( <code>screen.py</code> refuses unsourced attestations) | Equities screening (#370), IKA feed attestation (#570–#578), options exceptions (#637) all reuse one mechanism  |

| Decision (in the code today)                                                                                    | Carries                                                                                           |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <code>keel serve loopback</code> <b>by design</b> with DNS-rebinding bind checks                                | <code>keel link</code> (#648) extends a threat model that exists; security is not bolted on later |
| Release pipeline: 5 wheels + desktop matrix, gated signing, attestations; <code>keel update self-updater</code> | Pro-tier artifacts and managed updates are built and waiting, not engineered                      |
| Experiment records reproducible (#265); docs under test (incl. <code>test_workflow_triggers.py</code> )         | The Fee Reality benchmark's honesty (#646) is enforced by infrastructure                          |
| Evidence bound to the credential that earned it (#633/#635)                                                     | The audit primitive the hosted-confirm relay model requires                                       |

## 4. Security architecture

- **Loopback bind by design** with Host/bind checks against DNS rebinding; remote exposure is deliberately absent until #648 lands the opt-in bind, tunnel Host allowlist, and off-loopback token posture.
- **One-time session token** URLs; CSP `default-src 'self'` with no `unsafe-inline` (pinned by test); the key-parity scanner ensures the payload emits every key the client reads.
- **MCP is read-only at the engine level** (`query_only`): it cannot attest, promote, arm autonomy, or place an order — by construction, not policy.
- **Credentials:** `.env` today; OS keychain via `keyring` is the #437 wizard's plan. Evidence is fingerprint-bound to credentials (#633).
- **Honest refusal as a security property:** Robinhood refuses rather than faking; Kraken stubs loudly; installer truths are pinned by tests.

## 5. Data architecture

- SQLite WAL, one database per profile — ADR 0002 records why and the trigger that would change it. Migrations via `keel migrate`; idempotent-on-start is #437's ask.
- **The trials ledger** — hash-chained JSONL, schema v11; the trials budget with provenance. Append-only: dated records never rewritten.
- **Freshness is measured** (`keel/data/freshness.py`) — the open gap is operational: wrappers do not gate on it (`fetch` → `doctor` → `cycle` is the filed shape).
- **Spread is unmeasured** (#626) — at this deployment's clip sizes, spread is the actual cost; the measurement program is the live book's purpose.

## 6. Serving and interface architecture

- `keel/web`: `server.py` (bind + routing), `api.py` / `payload.py` (one JSON contract, key-parity pinned), `events.py` (SSE live updates), `staticfiles.py`, `security.py` (header set).

- The web UI is **zero-dependency** static ES modules — view-source is a feature. Its constraints are pinned by tests: identity pins, palette contrast suite, deterministic icon generation, the CSP one-script pin.
- PWA: manifest + service worker ship for offline/install only. **Web Push is greenfield** (corrected 2026-08-31, #649): no push handler, subscription endpoint, or payload encryption exists; all are new work behind #648.

## 7. Release and distribution architecture

release.yml (workflow\_dispatch): version-agreement tests → uv lock check → suite → tag → build wheels + desktop matrix (macOS DMGs, Windows zip + Inno Setup setup.exe) → SHA256SUMS + attestations → publish. **Signing is built but gated**: absent secrets produce honest skip notes (the #438 Option D decision — activation is a secrets-drop, not a project). CI runs on every merge by design (post-merge runs see semantic merge conflicts); docs-only changes skip the workflows that read no documents (#644/#645), while ci.yml stays unfiltered because the docs themselves are under test.

## 8. Verified open architecture work

| Gap                                                                                             | Issue                                                                                  |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Remote-exposure security pass (bind opt-in, tunnel Host validation, off-loopback tokens)        | #648 — prerequisite for all of Phase B                                                 |
| Approval protocol primitives ( idempotency_key + proposal_hash , expiry, non-TTY confirm route) | design recorded in docs/evolution-plan.md §5 Phase B; implementation unfiled           |
| Web Push (VAPID, subscription, RFC 8291, sw handler)                                            | greenfield, gated on #648                                                              |
| Operational freshness gating (fetch → doctor → cycle; per-product staleness gate)               | the live-deployment session's open question; freshness.py measures, wrappers don't ask |
| Equities fundamentals pipeline (SEC XBRL companyfacts , pinned quarterly snapshot)              | #370                                                                                   |
| Options: amend or hold the constitution                                                         | #636 discovery → #637 ADR 0005                                                         |
| Control plane, BTCPay rail, hosted-confirm signer (native app or WebCrypto/dumb-relay spike)    | #638, docs/evolution-plan.md §5 Phase F — all external to the engine                   |

## 9. Architectural refusals on record

- **No shared-Postgres multi-tenant rewrite** — it would sacrifice ADR 0002 and the identical-artifact property between tiers.
- **No engine feature gates for paid tiers** — the free engine is never a demo.
- **No feed-derived ranking keys** — an assessment may inform a human attestation; it may never be one (#570–#578 rails).
- **No affiliate revenue, no strategies for sale, no hosted black box** — recorded in ADR 0004's scope (#603) and docs/evolution-plan.md §2.

## 10. Traceability

- ADRs: `docs/decisions/` — 0001 desktop update path, 0002 SQLite persistence, 0003 commands-layer survey; 0004 (monetisation) pending #603; 0005 (options) pending #637.
- Strategy: `docs/evolution-plan.md`.
- Runbooks: `docs/operator-runbook.md`, `docs/go-live-runbook.md`, `docs/RELEASING.md`.
- This document's rule: a claim about the present is verified against `main` at the date above; a claim about the future names its issue.

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